



News & Views

The first decline in U.S. wind power generation despite the installed capacity growth in the 21st century

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Wind energy has become one of the central pillars of the United States' clean energy transition [1,2]. Over the past two decades, both the installed capacity of wind turbines and wind power generation have grown rapidly, supported by technological progress and supportive policies [3,4]. The share of wind power in national electricity generation has expanded from less than 1% in 2000 to over 10% by 2022, positioning the U.S. as the world's second-largest wind energy producer [5]. Continuous technological improvements, including taller towers and longer blades, have enabled the capture of weaker and more variable winds, reinforcing confidence in the sector's steady ascent [6,7]. Yet this long-standing trajectory was abruptly interrupted in 2023. For the first time since large-scale wind development began, U.S. wind power generation recorded a year-on-year decline, despite the newly installed capacity remaining high [8]. This unexpected reversal marks the sensitivity of national-scale wind power generation to inter-annual wind speed variability. Beyond its immediate surprise, the downturn underscores a fundamental vulnerability in wind energy that the wind power depends on the variability of wind itself. When large-scale atmospheric circulation patterns weaken wind speeds, even the most advanced turbines cannot sustain expected output levels.

Since the onset of large-scale wind deployment, both installed capacity and annual generation have shown a steady upward trend across the contiguous United States (CONUS). National wind capacity has continued to expand, while annual growth in wind power generation rose almost continuously over the past two decades (Fig. 1a). The year 2022 marked a record high, with the wind power generation growing by an impressive 56.1 TWh (Fig. 1b). However, the generation unexpectedly dropped by 9.1 TWh in 2023 (Fig. 1b), even as new capacity continued to be added (Fig. 1a). This rare decoupling between installed capacity and actual output underscores a fundamental truth that the physical resource that sustains wind power is inherently variable.

Wind power generation is proportional to the cube of wind speed [9], such that changes in near-surface wind speed can lead to pronounced variations in electricity production. Against this

physical backdrop, the anomalous decline in wind power generation in 2023 is plausibly linked to a weakening of the wind resource (Fig. 1c). Using nearly two decades of observations across CONUS as a climatological reference (baseline period: 2000–2021), annual mean wind speeds in 2023 fall statistically below the long-term average and outside the 95% confidence interval, indicating a statistically robust negative anomaly (Fig. 1c). Distributional analysis across more than 236 wind monitoring stations further corroborates this conclusion. When compared with the baseline distribution, the 2023 annual mean wind speeds exhibit a downward shift in both the median and mean values, confirming that the anomaly was spatially widespread rather than driven by a small subset of locations (Fig. 1d). In contrast, 2022 experienced anomalously strong winds relative to the baseline, with wind speeds reaching the highest levels observed since the post-2010 wind-speed reversal period. This unusually favorable wind regime likely amplified wind power generation in 2022, thereby accentuating the subsequent decline in 2023 when wind speeds dropped sharply. Taken together, the juxtaposition of an anomalously high-wind year followed by a statistically significant low-wind year underscores the sensitivity of national-scale wind power generation to inter-annual variability in wind resources.

Seasonal analysis further demonstrates that the 2023 downturn was neither temporally uniform nor limited to a short-lived anomaly, but instead reflected a persistent weakening of wind resources across multiple months relative to the long-term climatological baseline. When evaluated against the 2000–2021 monthly wind-speed climatology, 2023 exhibits statistically significant negative wind-speed anomalies across much of the year, including the core wind-production season, whereas 2022 is characterized by predominantly positive anomalies relative to the baseline (Fig. 1e). Consistent with this contrast in wind conditions, wind generation in 2023 was notably lower during April to July and again in November–December, with the most substantial monthly decrease occurring in May (−10.1 TWh) (Fig. 1f). The timing of these declines corresponds closely with observed reductions in wind speed (Fig. 1e). Mean wind speeds in 2023 were lower than in 2022 across most months, and the 95% confidence intervals indicate that this weakening was statistically robust across monitoring stations.

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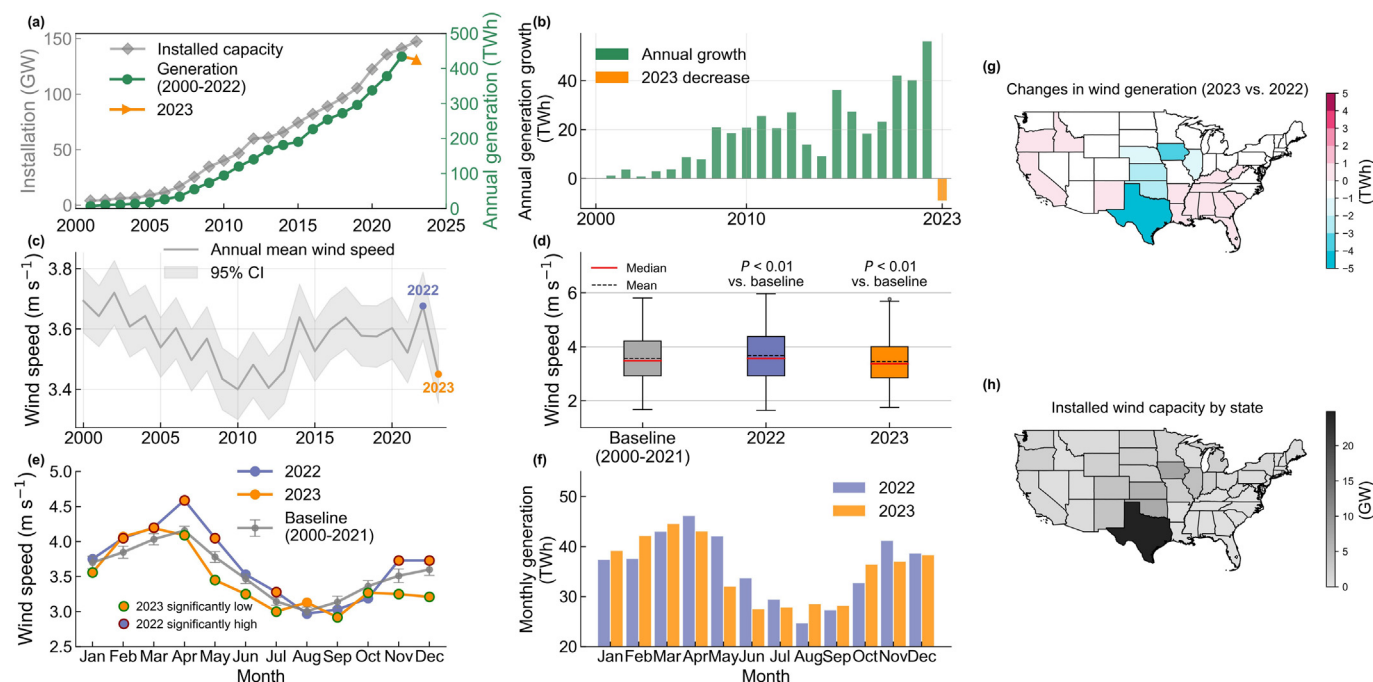


Fig. 1. Overview of wind speed and wind power generation changes in the contiguous United States (CONUS). (a) Cumulative installed wind power capacity (grey line) and total wind power generation (green line) in CONUS from 2000 to 2023. (b) Annual growth in wind power generation from 2000 to 2023. (c) Annual mean wind speed from 2000 to 2023 with 95% confidence interval (CI); wind speeds in 2022 (purple) and 2023 (orange) are highlighted. (d) Distribution of wind speeds during the baseline period (2000–2021; grey), 2022 (purple), and 2023 (orange), with the median (red solid line) and the mean (grey dashed line) indicated. (e) Monthly mean wind speed climatology for the baseline period (2000–2021; gray line with 95% CI) and monthly mean wind speeds in 2022 (purple) and 2023 (orange); months with significant positive and negative anomalies are indicated. (f) Comparison of seasonal wind power generation between 2022 and 2023. (g) Changes in wind power generation from 2022 to 2023. (h) Distribution of installed wind capacity by state in 2023.

Importantly, turbine availability metrics (including active, forced, and maintenance turbine hours) did not differ significantly from those in previous years (Fig. S1 online), suggesting that the downturn cannot be explained by operational disruptions or maintenance-related factors. Taken together, the concurrent and widespread reductions in both wind speed and wind generation point to a predominantly meteorological origin rather than operational or policy-related causes.

To further quantify the role of wind-speed variability, we adopt a previously established framework [10] that isolates the contribution of wind-speed changes to variations in wind power generation. Using this approach, we estimate that the reduction in wind speed in 2023 corresponds to an approximate decrease of 14.0 TWh in wind power generation across CONUS. This estimated wind-driven reduction is larger than the observed net decline of 9.1 TWh in total generation, indicating that the adverse impact of weakened winds was partially offset by ongoing technological improvements and continued capacity expansion. In other words, while unfavorable wind conditions exerted a substantial downward pressure on national wind generation, efficiency gains and newly installed capacity helped absorb part of this loss, resulting in a smaller net decline at the aggregate level.

Spatially, the impact of reduced winds was widespread but geographically uneven. The majority of U.S. states experienced declines in wind generation in 2023, particularly those located in the nation's wind-rich central corridor (Fig. 1g). Texas recorded the largest reduction (−4.82 TWh), followed by Iowa (−3.91 TWh), Oklahoma (−2.95 TWh), and Kansas (−2.27 TWh) (Table 1). In contrast, only a few states registered slight increases, all below 0.5 TWh, mostly in regions less reliant on wind power. These spatial contrasts emphasize the sensitivity of wind-rich states to interannual wind variability—a vulnerability that grows as their installed capacity expands.

Table 1

Representative U.S. states with the most pronounced decreases and increases in wind power generation in 2023.

State	Changes in wind generation vs. 2022 (TWh)	Cumulative wind power installation (GW)
Texas	−4.82	24.81
Iowa	−3.91	7.58
Oklahoma	−2.95	7.16
Kansas	−2.27	6.09
Oregon	+0.44	2.15
California	+0.33	1.06
New Mexico	+0.28	3.63

The distribution of installed capacity further reinforces this relationship (Fig. 1h). States with the largest installed capacity are also those most affected by the 2023 downturn, as their generation portfolios are heavily exposed to natural wind fluctuations. The observed mismatch between capacity growth and generation decline thus reflects not a failure of technology but a reminder of physical limits: when large-scale atmospheric circulation patterns weaken surface winds, even the most advanced turbines cannot fully compensate. As wind power continues to expand, understanding and anticipating such wind-speed variability will become increasingly important for ensuring grid stability, investment planning, and the long-term resilience of the renewable energy transition.

In conclusion, the first-ever decline in U.S. wind power generation highlights a critical intersection between renewable energy expansion and climate variability. The 2023 downturn does not diminish the long-term promise of wind, but it highlights a need to rethink resilience in renewable energy planning. This decline

event challenges the assumption that ongoing technological improvement alone can fully offset unfavorable climatic variability at the national scale. In the past, advances such as taller towers and longer blades have enhanced turbine efficiency, partially compensating for weaker winds. However, because wind power generation scales approximately with the cube of wind speed, even modest wind-speed reductions can translate into disproportionate losses in electricity production. As a result, even with record-high efficiency and capacity, the 2023 decline demonstrates that large-scale reductions in wind speed can still dominate production outcomes and lead to significant absolute generation losses. Regional patterns provide further insight into this vulnerability. The 2023 decline was geographically uneven, with the largest absolute generation losses concentrated in wind-rich central states. This spatial alignment between capacity concentration and generation decline underscores that regions most successful in scaling up wind deployment are also those most exposed to inter-annual wind variability. As wind power becomes a larger component of the electricity mix, its aggregate sensitivity to these fluctuations increases, transforming natural variability from a local to a system-wide concern.

Importantly, the observed divergence between continued growth in installed capacity and declining annual generation in 2023 does not necessarily indicate a saturation of wind power deployment or widespread turbine underperformance. Instead, our analysis suggests that this decoupling primarily reflects the exposure of an increasingly capacity-intensive system to unfavorable wind conditions. While wind performance in 2023 was lower than in some recent years, it remained within the historical range (Fig. S2 online), indicating that the decline was driven by meteorological variability rather than by fundamental limitations of turbine technology or site suitability. This highlights that, under changing climate conditions, even well-sited and technologically advanced wind fleets remain constrained by physical wind resources at large scales.

Looking ahead, the unprecedented downturn of 2023 should thus be viewed as a valuable lesson rather than a setback. As wind power accounts for an increasing share of the electricity mix, inter-annual wind speed fluctuations are increasingly translated into amplified system-wide impacts, underscoring the growing importance of managing climate variability. Sustaining growth in the era of climate uncertainty will require not only more turbines, but also smarter strategies that adapt to the shifting patterns of the atmosphere. At the planning level, energy agencies and policymakers should incorporate inter-annual and decadal wind variability into capacity planning, long-term generation forecasting, and risk assessment models, rather than relying solely on climatological mean conditions. At the system level, greater spatial diversification of wind projects can help reduce correlated low-wind risks across regions, while integration with complementary resources, such as solar generation and energy storage, can help buffer large-scale production anomalies and further enhance system resilience. At the operational level, improving meteorological forecasting and early-warning systems for low-wind conditions could help grid

operators and market regulators anticipate and mitigate potential shortfalls. Together, these strategies emphasize that sustaining growth in wind energy will require not only continued expansion of installed capacity, but also planning and operational frameworks that are explicitly designed to account for inter-annual variability in wind resources.

Conflict of interest

The authors declare they have no conflict of interest.

Acknowledgments

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scib.2026.02.045>.

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